

Notes: Chen, De, Hu, and Hwang (RFS, 2014)

Wisdom of Crowds: The Value of Stock Opinions Transmitted Through Social Media

Contents

1 Big picture

1.1	Question
1.2	Setting
1.3	Core challenge
1.4	Main contribution
1.5	Main result
1.6	Economic takeaway

2 Data and measurement

2.1	Seeking Alpha articles and comments
2.2	Traditional information sources
2.3	Textual tone variables
2.3.1	Article negativity
2.3.2	Comment negativity
2.3.3	Traditional news negativity
2.4	Abnormal returns
2.5	Earnings-surprise variable
2.6	Author-level variables
2.6.1	Author consistency
2.6.2	Disagreement
2.7	Descriptive facts

3 Models and empirical specifications

3.1	Baseline return regression
3.2	Calendar-time portfolio test
3.3	Return horizons
3.4	Number-of-comments interaction
3.5	Earnings-surprise regression
3.6	Articles about earnings
3.7	Author following and track record
3.8	Within-author popularity update
3.9	Author-reader disagreement
3.10	Poor-track-record correction test

4 Identification and credibility

4.1	What identifies the main estimate?
4.2	Why the first-two-day skip matters
4.3	Traditional information controls
4.4	Earnings surprises as a credibility test
4.5	No reversal and longer-horizon evidence
4.6	Crowd aggregation
4.7	Reputation and correction
4.8	Remaining limitations

5 Tables and figures

5.1	Table 1: Descriptive statistics of Seeking Alpha and DJNS content
5.2	Tables 2 and 3: Return variables and firm characteristics
5.3	Table 4: Seeking Alpha tone and abnormal returns
5.4	Figure 3: Return predictability over different horizons
5.5	Figure 4: Calendar-time trading strategy
5.6	Table 5: Number of comments
5.7	Table 6: Seeking Alpha tone and earnings surprises
5.8	Table 7: Author track record and following
5.9	Table 8: Author track record and follower disagreement

5.10	Table 9: Poor author track records and crowd correction	
6	Short summary	
7	Conclusion	

1 Big picture

1.1 Question

The paper asks whether investor opinions transmitted through social media contain value-relevant information. The empirical setting is Seeking Alpha, an online platform where investors write stock-focused articles and where readers respond with comments. The central question is simple:

Do peer investors, writing outside the traditional sell-side analyst and news-media system, produce opinions that predict future stock returns and future earnings news?

The paper is not merely asking whether social media moves prices immediately. It asks whether the tone of social-media stock opinions contains information that is later confirmed by returns and earnings surprises.

1.2 Setting

Seeking Alpha is a useful setting because it is neither a short-message chat room nor a purely professional research platform. Articles are relatively long, are reviewed by editors, are tied to specific stocks, and generate a public comment section. This lets the authors observe two related but distinct signals:

- **Author signal:** the tone of the Seeking Alpha article.
- **Crowd signal:** the tone of comments written in response to the article.

The authors compare these signals to traditional information sources, including Dow Jones News Service articles, sell-side analyst recommendation changes, and earnings surprises.

1.3 Core challenge

The empirical difficulty is that social media can be interpreted in several ways:

- **Information channel:** articles and comments reveal information that prices do not yet fully incorporate.
- **Naive-reaction channel:** investors trade on noisy or misleading opinions and slowly push prices in that direction.
- **Random-chatter channel:** social media contains little useful information and any return relation is spurious.

The paper's design tries to separate these views by skipping the first two days after publication, looking at medium-run returns, testing for earnings-surprise predictability, and studying how author reputation and comment disagreement affect informativeness.

1.4 Main contribution

The paper contributes to the literature in four main ways.

1. It studies a large social-media investment platform rather than small message-board samples.
2. It separates article tone from comment tone, allowing the author signal and reader signal to be compared.
3. It shows that social-media tone predicts not only future returns, but also future earnings surprises.
4. It studies mechanisms: whether the crowd aggregates information, whether author track records affect attention, and whether comments can correct weak authors.

1.5 Main result

Negative tone in Seeking Alpha articles predicts lower future abnormal returns. Negative tone in the comments also predicts lower future abnormal returns. In the baseline return regression, the coefficient on article negativity is about -0.320 , and the coefficient on comment negativity is about -0.196 when traditional news controls are included.

Interpretation. A one-percentage-point increase in the fraction of negative words in the article is associated with about 0.320 percentage points lower abnormal returns over the next three months. A one-percentage-point increase in the fraction of negative words in the comments is associated with about 0.196 percentage points lower abnormal returns over the same horizon.

1.6 Economic takeaway

The paper's main takeaway is that investor-generated research on social media is not just noise. At least in this setting, peer opinions contain information that is later reflected in prices and earnings. The comment section is especially important because it can aggregate the views of multiple readers and, in some cases, challenge or correct the article author.

2 Data and measurement

2.1 Seeking Alpha articles and comments

The authors download Seeking Alpha articles from 2005 to 2012. They focus on **single-ticker articles**, because for these articles the discussed stock is clear. Multiple-ticker articles are excluded because it is difficult to assign article tone to one specific company.

Key sample objects:

- **SA articles:** 97,070 single-ticker articles.
- **SA comments:** 459,679 comments posted within the first two days after article publication.
- **Coverage:** more than 7,000 firms in the broader article sample.

- **Users:** roughly 6,500 article authors and about 180,000 comment writers over the sample period.

The first-two-day comment window matters because roughly 80% of comments arrive on the publication day or the following day. Later comments are more sporadic and may be less connected to the article’s original information event.

2.2 Traditional information sources

The paper also uses:

- **Dow Jones News Service (DJNS):** to measure the tone of traditional news coverage.
- **IBES recommendations:** to control for analyst upgrades and downgrades.
- **IBES earnings forecasts:** to compute earnings surprises.
- **CRSP:** to measure stock returns.
- **Compustat:** to obtain accounting and firm-level variables.

The key idea is to ask whether Seeking Alpha adds information beyond standard analyst and news channels.

2.3 Textual tone variables

The paper measures tone using the Loughran–McDonald negative-word dictionary. The main tone variables are fractions of negative words.

2.3.1 Article negativity

$$NegSA_{i,t} = \frac{\text{number of negative words in SA articles about firm } i \text{ on day } t}{\text{total number of words in those SA articles}}.$$

Interpretation. Higher $NegSA_{i,t}$ means that the article-level discussion of firm i is more negative on day t .

2.3.2 Comment negativity

$$NegSA-Comment_{i,t} = \frac{\text{number of negative words in comments on SA articles about firm } i}{\text{total number of words in those comments}},$$

where comments are measured over days t to $t + 1$ after article publication.

Interpretation. This is the tone of the reader response. If the article author is optimistic but the comments are negative, this variable captures the crowd’s disagreement with the author.

2.3.3 Traditional news negativity

$$NegDJNS_{i,t} = \frac{\text{number of negative words in DJNS articles about firm } i \text{ on day } t}{\text{total number of words in those DJNS articles}}.$$

Interpretation. This variable is included to show that the Seeking Alpha effect is not simply repackaged traditional news sentiment.

2.4 Abnormal returns

The dependent variable in the main return tests is a three-month characteristic-adjusted abnormal return:

$$ARet_{i,t+3,t+60}.$$

The abnormal return is the firm’s raw return minus the return on a value-weighted portfolio of firms with similar size, book-to-market ratio, and past returns, following the Daniel–Grinblatt–Titman–Wermers style of characteristic adjustment.

The holding period starts at $t + 3$ and ends at $t + 60$.

Interpretation. The authors skip the first two days after article publication so that the return window does not overlap with the period used to compute article and comment tone. This helps reduce the concern that the results are driven only by immediate price pressure around publication.

2.5 Earnings-surprise variable

For the earnings tests, the dependent variable is a price-scaled earnings surprise:

$$ES_{i,t} = \frac{EPS_{i,t}^{reported} - EPS_{i,t}^{forecast}}{Price_{i,t}}.$$

The forecast is the average analyst EPS forecast issued or updated within 30 days before the earnings announcement.

Interpretation. If Seeking Alpha tone predicts future earnings surprises, the paper has stronger evidence that social-media opinions contain value-relevant information, because earnings outcomes are unlikely to be caused by social-media posts.

2.6 Author-level variables

The paper also studies reputation and crowd correction.

2.6.1 Author consistency

For each author, the authors classify each prior article as bullish or bearish based on its negative-word share. An article is **consistent** if:

- a bullish article is followed by positive abnormal returns; or
- a bearish article is followed by negative abnormal returns.

Then:

$$Consistency_i =$$

racnumber of consistent prior articles by author i number of prior single-ticker articles by author i.

Interpretation. $Consistency_i$ is a track-record measure. Higher values mean that an author’s past article tone has more often aligned with subsequent stock performance.

2.6.2 Disagreement

For an article j by author i :

$$Disagreement_{i,j} = |NegSA_{i,j} - NegSA-Comment_{i,j}|.$$

The author-level disagreement measure averages this object across articles.

Interpretation. Disagreement is high when the author and the comment crowd express very different tones. This is useful for studying whether the crowd challenges less reliable authors.

2.7 Descriptive facts

The descriptive evidence establishes that the textual data are substantial rather than trivial:

- The average Seeking Alpha article has about 675 words.
- The average Seeking Alpha comment has about 82 words.
- The average DJNS article has about 380 words.
- The average negative-word share is about 1.25% in SA articles, 1.75% in SA comments, and 1.48% in DJNS articles.
- The main return regression uses 40,946 firm-day observations.

A key descriptive point is that Seeking Alpha content resembles informal analyst research more than short message-board chatter. This helps explain why the paper finds stronger predictability than earlier message-board studies.

3 Models and empirical specifications

3.1 Baseline return regression

The central regression is:

$$ARet_{i,t+3,t+60} = \alpha + \beta_1 NegSA_{i,t} + \beta_2 NegSA-Comment_{i,t} + X_{i,t}\delta + \varepsilon_{i,t}. \quad (1)$$

$X_{i,t}$ includes controls for whether there are SA comments, DJNS tone, whether there is DJNS coverage, analyst upgrades and downgrades, positive and negative earnings surprises, volatility, short-run past abnormal returns, and year-month fixed effects. Standard errors are clustered by firm and year-month.

Interpretation. The coefficient β_1 asks whether the author’s negative tone predicts future abnormal returns. The coefficient β_2 asks whether the readers’ negative tone predicts future abnormal returns after controlling for the article author’s tone.

3.2 Calendar-time portfolio test

The paper also forms portfolios based on tone:

1. At the end of day t , sort stocks into tone portfolios using $NegSA_{i,t}$ or $NegSA-Comment_{i,t}$.
2. Skip two days.
3. Hold each stock for three months.
4. Go long the least-negative tone portfolio and short the most-negative tone portfolio.

Interpretation. This converts the regression signal into a portfolio object. If the signal is economically meaningful, the long-short portfolio should earn positive abnormal returns over time.

3.3 Return horizons

The paper repeats the baseline return regression using multiple horizons:

1 month, 3 months, 6 months, 12 months, 36 months.

Interpretation. If negative tone merely causes temporary price pressure, the effect should reverse or disappear quickly. If it reflects value-relevant information, the effect can persist or accumulate over longer horizons.

3.4 Number-of-comments interaction

To test whether the comment signal is stronger when it aggregates more readers, the paper estimates a specification of the form:

$$ARet_{i,t+3,t+60} = \alpha + \beta_1 NegSA_{i,t} + \beta_2 NegSA-Comment_{i,t} + \beta_3 (NegSA-Comment_{i,t} \times Rank(\#Comment_{i,t})) + \gamma Rank(\#Comment_{i,t})$$

The comment-count rank is a tercile variable equal to 0, 1, or 2.

Interpretation. A negative β_3 means that comment tone becomes more predictive when more comments are available. This is a crowd-aggregation result: the crowd signal is stronger when it is based on more inputs.

3.5 Earnings-surprise regression

The earnings-surprise test can be summarized as:

$$ES_{i,t} = \alpha + \beta_1 NegSA_{i,t-30,t-3} + \beta_2 NegSA-Comment_{i,t-30,t-3} + \beta_3 NegDJNS_{i,t-30,t-3} + W_{i,t} \delta + \eta_{i,t} + \varepsilon_{i,t}.$$

Here, the text variables are measured from 30 days to three days before the earnings announcement. Controls include lagged earnings surprise, forecast dispersion, firm size, market-to-book, and recent returns.

Interpretation. This regression asks whether social-media tone predicts future fundamentals after controlling for analysts and traditional news. This is the paper’s most important test against the pure price-pressure interpretation.

3.6 Articles about earnings

The paper splits article and comment tone according to whether the underlying article contains the word stem “earn.” This produces variables such as:

$$NegSA_EA_{i,t-30,t-3}, \quad NegSA_NoEA_{i,t-30,t-3}.$$

Interpretation. If the predictability is connected to earnings information, then articles related to earnings should have stronger earnings-surprise predictability than articles not related to earnings.

3.7 Author following and track record

The author-level mechanism test is:

$$Y_i = \alpha + \beta Consistency_i + Z_i \gamma + u_i,$$

where Y_i is either page views or read-to-end counts for articles written in the second half of 2012. Controls include article length, article negativity, whether the author has a blog, and whether the author discloses a company affiliation.

Interpretation. If readers can distinguish strong authors from weak authors, then authors with better past consistency should receive more attention.

3.8 Within-author popularity update

The paper also examines changes in popularity within an author:

$$\Delta Y_{i,j} = \alpha + \beta_1 ConsistencyRecentArticles_{i,j-1} + \varepsilon_{i,j}. \quad (2)$$

Interpretation. This asks whether an author’s recent successful calls increase later attention. It is a dynamic reputation test.

3.9 Author-reader disagreement

The disagreement regression is:

$$Disagreement_i = \alpha + \beta Consistency_i + Z_i\gamma + u_i.$$

Interpretation. A negative β means that readers disagree less with authors who have better historical records and more with authors who have worse historical records.

3.10 Poor-track-record correction test

The final mechanism test returns to the return regression but restricts the sample to cases where:

1. the author and comments disagree; and
2. the author has a poor past track record.

Interpretation. If the comment crowd is corrective, then in this restricted sample the comment tone should be more reliable than the article tone.

4 Identification and credibility

4.1 What identifies the main estimate?

The main variation comes from differences in textual tone across firm-day Seeking Alpha coverage. Conditional on the controls and year-month fixed effects, the paper asks whether more negative tone predicts lower future abnormal returns or lower future earnings surprises.

This is predictive evidence rather than a randomized experiment. The paper is careful about this distinction: it does not claim that Seeking Alpha randomly causes information to appear. Instead, it shows that the observed text contains information that predicts later outcomes.

4.2 Why the first-two-day skip matters

The return window begins at $t + 3$, while article tone is measured on day t and comment tone over days t to $t + 1$.

Interpretation. This design separates measurement of tone from measurement of returns. It reduces the concern that the results are driven by immediate trading on the publication day or by mechanical overlap between the signal window and the return window.

4.3 Traditional information controls

The paper controls for analyst upgrades and downgrades, earnings-surprise indicators, DJNS tone, whether there is DJNS coverage, volatility, and recent abnormal returns.

Interpretation. These controls are important because Seeking Alpha posts may discuss the same

news that analysts or traditional news outlets already cover. The result is stronger because SA article and comment tone continue to predict returns after these controls are included.

4.4 Earnings surprises as a credibility test

The earnings-surprise test is central. If negative Seeking Alpha tone predicts lower future earnings surprises, then the tone is not only associated with future prices; it is associated with future fundamentals.

Interpretation. This makes the pure naive-investor-reaction story less persuasive. Social-media comments may influence trading, but they are unlikely to cause quarterly earnings outcomes.

4.5 No reversal and longer-horizon evidence

Figure 3 shows that return predictability does not simply reverse quickly. Coefficients generally remain negative over several horizons.

Interpretation. A temporary-price-pressure story would predict a reversal after initial attention fades. The absence of a strong reversal is more consistent with gradual incorporation of information.

4.6 Crowd aggregation

The comment signal is stronger when it is based on more comments.

Interpretation. This is exactly what one would expect if comments aggregate dispersed information. A noisy single comment is weak, but a collection of comments can contain a stronger signal.

4.7 Reputation and correction

The paper finds that authors with better historical consistency receive more page views and read-to-end counts. It also finds that readers disagree less with historically consistent authors, and that when readers challenge poor-track-record authors, the comments are more predictive than the article.

Interpretation. These results support a mechanism in which readers learn about author quality and comments can discipline or correct weak authors.

4.8 Remaining limitations

Important caveats remain:

- The paper does not have random assignment of article tone or reader attention.
- Textual negativity is an imperfect proxy for investor opinion.
- Seeking Alpha editorial screening may create selection into published articles.
- Authors may choose which firms to cover based on private or public signals not fully observed

Table 1
Descriptive statistics of Seeking Alpha and Dow Jones News Service articles

	Seeking Alpha (SA) Articles	Seeking Alpha (SA) Comments	Dow Jones News Service (DJNS) Articles
Panel A: Single-Ticker SA Articles, SA Comments and DJNS Articles			
Total # Stock tickers	7,422	5,031	4,507
Total # Articles (or Comments)	97,070	459,679	322,046
Avg. # Words per article	675	82	380
StDev. # Words per article	466	104	934
Avg. % Negative words	1.25%	1.75%	1.48%
StDev % Negative words	0.96%	2.74%	1.49%
Panel B: SA and DJNS Articles with Word Stem "Earn" and Corresponding SA Comments			
Total # Stock tickers	5,054	3,406	3,889
Total # Articles	45,239	200,546	100,403
Avg. # Words per article	741	79	455
StDev. # Words per article	520	101	836
Avg. % Negative words	1.20%	1.63%	1.49%
StDev % Negative words	0.88%	2.62%	1.20%

This table reports summary statistics for single-ticker *Seeking Alpha* (SA) articles, SA comments written in response to single-ticker SA articles and *Dow Jones News Service* (DJNS) articles. In Panel B, we report summary statistics for the subset of single-ticker SA articles with the word stem "earn" and the SA comments written in response to these SA articles, as well as DJNS articles with the word stem "earn." The sample period is 2005–2012.

by the econometrician.

- The trading-strategy results do not fully account for trading costs, short-sale constraints, or implementation frictions.

These limitations do not overturn the evidence, but they shape the interpretation. The safest reading is that social-media tone is a meaningful predictive signal, not that the paper fully identifies a causal effect of social media on prices.

5 Tables and figures

5.1 Table 1: Descriptive statistics of Seeking Alpha and DJNS content

Read: Table 1 reports the scale and tone of the text data. Seeking Alpha articles are longer than DJNS articles on average, comments are shorter but still meaningful, and the negative-word shares are comparable across SA articles, SA comments, and DJNS articles.

Economic message: The dataset is large enough and rich enough to study investor opinion. The comments are not just one-word reactions; their average length is substantially longer than typical message-board posts studied in earlier work.

5.2 Tables 2 and 3: Return variables and firm characteristics

Read: Table 2 gives summary statistics for firm-day variables used in return regressions. Table 3 shows that sample firms are larger, more growth-oriented, have higher past returns, and have more analyst coverage than the average CRSP/Compustat firm.

Economic message: Seeking Alpha coverage is not randomly assigned across all firms. The sample tilts toward firms that investors are more likely to discuss. This makes controls and characteristic-adjusted abnormal returns important.

Table 2
Summary statistics: Firm/trading day level

	<i>N</i>	Mean	Std. Dev.	25 th Pctl	50 th Pctl	75 th Pctl
$ARet_{i,t+3,t+60}$	40,946	-0.004	0.162	-0.079	-0.006	0.067
$ARet_{i,t}$	40,946	0.000	0.040	-0.010	0.000	0.010
$ARet_{i,t-1}$	40,946	0.000	0.041	-0.010	0.000	0.010
$ARet_{i,t-2}$	40,946	0.000	0.031	-0.009	0.000	0.009
$ARet_{i,t-60,t-3}$	40,946	0.004	0.241	-0.086	-0.008	0.073
$Upgrade_{i,t}$	40,946	0.030	0.167	0.000	0.000	0.000
$Downgrade_{i,t}$	40,946	0.034	0.180	0.000	0.000	0.000
$PosES_{i,t}$	40,946	0.060	0.238	0.000	0.000	0.000
$NegES_{i,t}$	40,946	0.016	0.126	0.000	0.000	0.000
$Volatility_{i,t}$	40,946	0.018	0.059	0.004	0.008	0.017

This table reports the summary statistics of the main variables. The observations are on a firm-day level. Abnormal returns ($ARet_{i,t}$) are company i 's raw returns minus the return of a value-weighted portfolio with similar size/book-to-market/past-return characteristics; t is the day an article about company i is published on the *Seeking Alpha* website, or the ensuing trading day if the article is published on a non-trading day. $Upgrade_{i,t}$ and $Downgrade_{i,t}$ are the number of financial analysts upgrading and downgrading company i on day t . $PosES_{i,t}$ and $NegES_{i,t}$ are indicator variables denoting whether company i experienced a positive (negative) earnings surprise on day t . $Volatility_{i,t}$ is the sum of squared daily returns in the calendar month prior to day t .

Table 3
Summary statistics: Firm/calendar year level

	<i>N</i>	Mean	Std. Dev.	25 th Pctl	50 th Pctl	75 th Pctl
<i>Size</i>	7,773	10,291	29,424	529	1,930	7,204
<i>BM</i>	7,773	0.640	1.080	0.274	0.470	0.760
<i>Past Return</i>	7,773	0.140	1.300	-0.200	0.070	0.310
<i>Coverage</i>	7,773	10.870	7.840	5.000	10.000	16.000
<i>Retail Holdings</i>	7,773	0.260	0.230	0.083	0.210	0.390

This table reports the summary statistics of various firm characteristics. The observations are on a firm-year level. Every year t (from 2005 to 2012), we compile a list of firms in our sample. We then compute the respective firms' characteristics as of December. *Size* is the firm's market capitalization in millions. *BM* is the firm's book-to-market ratio. *Past Return* is the firm's cumulative one-year return. *Coverage* is the firm's analyst coverage, which is set equal to zero if the firm is not covered by any analysts in that year. *Retail Holdings* is one minus the fraction of shares held by institutional investors.

5.3 Table 4: Seeking Alpha tone and abnormal returns

Read: Table 4 regresses three-month abnormal returns on article negativity and comment negativity. Column (3), the fullest specification, includes DJNS controls and other firm-day controls. The coefficient on $NegSA_{i,t}$ is -0.320 , and the coefficient on $NegSA-Comment_{i,t}$ is -0.196 .

Economic message: Both the author signal and the comment signal are predictive. Importantly, the comment signal remains useful after controlling for article tone. This means that reader reactions add information beyond the original article.

Table 4
Seeking Alpha and abnormal returns

	(1)	(2)	(3)
$NegSA_{i,t}$	-0.379 (-2.24)	-0.332 (-2.03)	-0.320 (-1.98)
$NegSA-Comment_{i,t}$		-0.194 (-3.44)	-0.196 (-3.55)
$I(SA-Comment_{i,t})$		0.001 (0.25)	0.001 (0.17)
$NegDJNS_{i,t}$			-0.254 (-1.44)
$I(DJNS_{i,t})$			0.009 (1.33)
$Upgrade_{i,t}$	0.003 (0.59)	0.003 (0.60)	0.003 (0.50)
$Downgrade_{i,t}$	-0.005 (-1.08)	-0.005 (-1.06)	-0.005 (-1.10)
$PosES_{i,t}$	0.0014 (0.38)	0.001 (0.25)	-0.002 (-0.41)
$NegES_{i,t}$	-0.004 (-0.44)	-0.004 (-0.49)	-0.006 (-0.66)
$Volatility_{i,t}$	-0.044 (-0.52)	-0.043 (-0.50)	-0.042 (-0.49)
$ARet_{i,t}$	-0.068 (-1.64)	-0.070 (-1.68)	-0.071 (-1.71)
$ARet_{i,t-1}$	-0.077 (-2.00)	-0.077 (-2.00)	-0.077 (-2.01)
$ARet_{i,t-2}$	-0.021 (0.35)	-0.022 (0.37)	-0.022 (-0.38)
$ARet_{i,t-60,t-3}$	-0.021 (-1.41)	-0.022 (-1.42)	-0.022 (-1.43)
# Obs.	40,946	40,946	40,946
Adj. R^2	1.20%	1.23%	1.24%

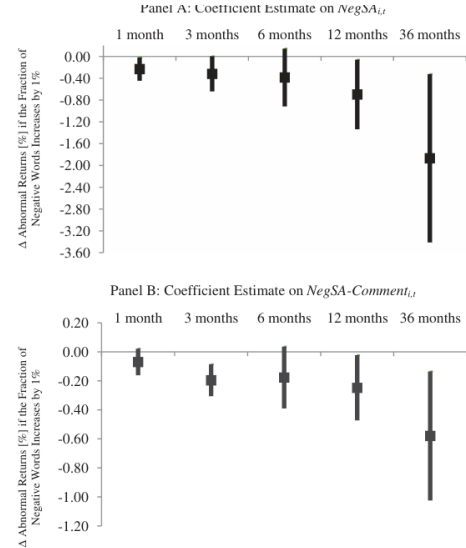


Figure 3
Seeking Alpha and abnormal returns over different holding periods

This figure reports coefficient estimates from regressions of abnormal returns on measures of the views reflected in *Seeking Alpha* (SA) articles and comments. The sample period is 2005–2012. Abnormal returns are company's raw returns minus the return of a value-weighted portfolio with similar size/book-to-market/past-return characteristics. The horizons over which cumulative abnormal returns are computed are 1 month, 3 months, 6 months, 12 months, and 36 months. The regression equation is identical to the one in column (3) of Table 4. Here, we plot the coefficient estimates on $NegSA_{i,t}$ and $NegSA-Comment_{i,t}$ along with their corresponding 95% confidence intervals. Standard errors are clustered by firm and year-month.

5.4 Figure 3: Return predictability over different horizons

Read: Figure 3 plots coefficients on article negativity and comment negativity for return horizons of one month, three months, six months, one year, and three years. The coefficients are generally negative and often statistically different from zero.

Economic message: The signal is not only an immediate publication-day effect. The negative tone appears to forecast medium-run and longer-run underperformance. This supports a gradual-information-incorporation interpretation.

5.5 Figure 4: Calendar-time trading strategy

Read: Figure 4 shows the value of one dollar invested in long-short portfolios that go long less-negative stocks and short more-negative stocks, using either article tone or comment tone.

Economic message: The return signal is economically meaningful over time. The trading pattern is not confined to a single brief episode, although performance is stronger in some periods than others.

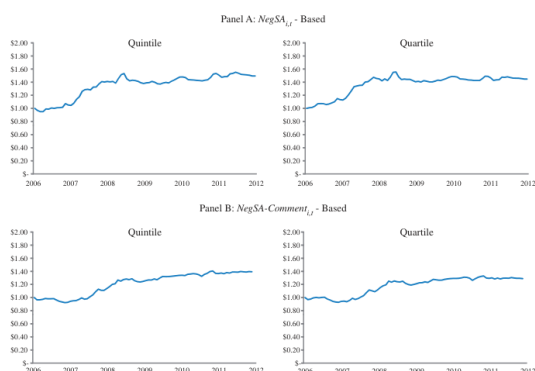


Figure 4
Calendar-time trading strategy based on Seeking Alpha
This figure depicts how \$1 invested in a simple calendar-time trading strategy would have evolved. The trading strategy is as follows: At the end of each trading day t , we assign stocks into quintile (quartile) portfolios based on the average fraction of negative words across all articles published on SA about company i on day t ($NegSA_{i,t}$); we also form quintile (quartile) portfolios based on the average fraction of negative words across SA comments posted over days t to $t+1$ in response to the SA articles ($NegSA-Comment_{i,t}$). We skip two days and hold each stock in its respective portfolio for three months. Based on the daily returns of a long-short portfolio, where we go long stocks in the bottom quintile (quartile) and short stocks in the top quintile (quartile), we plot how much \$1 would have grown/shrunk through calendar time.

Table 5
Seeking Alpha, abnormal returns, and number of Seeking Alpha comments

	(1)	(2)
$NegSA_{i,t}$	-0.393 (-1.95)	-0.381 (-1.92)
$NegSA-Comment_{i,t}$	-0.120 (-1.74)	-0.122 (-1.77)
$NegSA-Comment_{i,t} * Rank(\#SA-Comment_{i,t})$	-0.196 (-2.18)	-0.196 (-2.21)
$Rank(\#SA-Comment_{i,t})$	0.009 (1.93)	0.009 (2.05)
$NegDJNS_{i,t}$		-0.226 (-1.05)
$I(DJNS_{i,t})$		0.007 (0.98)
$Upgrade_{i,t}$	0.007 (0.92)	0.006 (0.84)
$Downgrade_{i,t}$	-0.006 (-0.84)	-0.006 (-0.84)
$PosES_{i,t}$	0.003 (0.59)	0.000 (0.01)
$NegES_{i,t}$	-0.013 (-0.79)	-0.015 (-0.82)
$Volatility_{i,t}$	-0.037 (-0.40)	-0.036 (-0.40)
$ARet_{i,t}$	-0.118 (-2.13)	-0.120 (-2.15)
$ARet_{i,t-1}$	-0.071 (-1.39)	-0.071 (-1.40)
$ARet_{i,t-2}$	-0.103 (-1.38)	-0.104 (-1.38)
$ARet_{i,t-60,t-3}$	-0.031 (-1.70)	-0.031 (-1.71)
# Obs.	21,124	21,124
Adj. R^2	2.26%	2.27%

This table reports coefficient estimates from regressions of abnormal returns on measures of the views reflected in Seeking Alpha (SA) articles and comments. The sample period is 2005–2012. The sample consists of observations with SA comments. Abnormal returns are the company's raw returns minus the return of a value-weighted portfolio with similar size/book-to-market/past return-characteristics from $t+3$ to $t+60$, where t is the day of article appearance or the ensuing trading day if the article is published on a non-trading day. $NegSA_{i,t}$ is the average fraction of negative words across all articles published on SA about company i on day t . $NegSA-Comment_{i,t}$ is the average fraction of negative words across SA comments posted over days t to $t+1$ in response to the SA articles, if there were any such comments, and zero otherwise. $Rank(\#SA-Comment_{i,t})$ is the tercile rank of number of SA-comments posted; realizations of $Rank(\#SA-Comment_{i,t})$ either equal zero, one, or two. $NegDJNS_{i,t}$ is the average fraction of negative words across all articles published in the DJNS about company i on day t , if there were any such articles, and zero otherwise. $I(SA-Comment_{i,t})$ and $I(DJNS_{i,t})$ are indicator variables denoting whether there were comments posted on SA articles and whether there were articles published in the DJNS. We include year-month fixed effects. Other independent variables are as described in Table 2. T -statistics are computed using standard errors clustered by firm and year-month and are reported in parentheses.

5.6 Table 5: Number of comments

Read: Table 5 interacts comment negativity with a tercile rank of the number of comments. The interaction term is negative and statistically significant. The implied effect of a one-percentage-point increase in comment negativity is much larger in the top comment-count tercile than in the bottom tercile.

Economic message: Comment tone is more informative when it is based on more comments. This is a key wisdom-of-crowds result: aggregation helps convert individual opinions into a stronger signal.

5.7 Table 6: Seeking Alpha tone and earnings surprises

Read: Table 6 regresses price-scaled earnings surprises on article and comment negativity measured from 30 days to three days before earnings announcements. Article negativity is strongly negative. Comment negativity is also negative, and the predictability is stronger when the article contains the word stem “earn.”

Economic message: This table is the strongest evidence that Seeking Alpha opinions are value relevant. Earnings surprises are not plausibly caused by SA comments, so the predictive relation points toward information rather than pure investor reaction.

Table 6
Seeking Alpha and earnings surprises

	(1)	(2)	(3)	(4)	(5)
$NegSA_{i,t-30,t-3}$	-0.266 (-2.45)	-0.232 (-2.27)			
$NegSA_EA_{i,t-30,t-3}$			-0.306 (-2.54)	-0.267 (-2.34)	-0.229 (-1.72)
$I(NegSA_EA_{i,t-30,t-3})$			0.001 (0.72)	0.001 (0.64)	0.005 (1.62)
$NegSA_NoEA_{i,t-30,t-3}$			-0.209 (-1.48)	-0.193 (-1.43)	0.020 (0.18)
$I(NegSA_NoEA_{i,t-30,t-3})$			0.000 (0.15)	0.001 (0.36)	0.002 (0.61)
$NegSA_Comment_{i,t-30,t-3}$	-0.095 (-1.72)	-0.094 (-1.72)			
$I(SA_Comment_{i,t-30,t-3})$	-0.002 (-1.43)	-0.002 (-1.31)			
$NegSA_Comment_EA_{i,t-30,t-3}$			-0.146 (-2.25)	-0.144 (-2.28)	-0.100 (-1.59)
$I(SA_Comment_EA_{i,t-30,t-3})$			0.000 (0.45)	0.000 (0.49)	0.000 (0.05)
$NegSA_Comment_NoEA_{i,t-30,t-3}$			-0.023 (-0.30)	-0.019 (-0.25)	-0.008 (-0.08)
$I(SA_Comment_NoEA_{i,t-30,t-3})$			-0.003 (-1.75)	-0.003 (-1.66)	0.001 (0.25)
$NegDINS_{i,t-30,t-3}$		-0.113 (-1.52)			
$I(DINS_{i,t-30,t-3})$		-0.001 (-0.57)			
$NegDINS_EA_{i,t-30,t-3}$				-0.127 (-1.68)	-0.073 (-1.50)
$I(DINS_EA_{i,t-30,t-3})$				0.001 (0.78)	0.001 (0.99)
$NegDINS_NoEA_{i,t-30,t-3}$				-0.039 (-0.63)	-0.110 (-0.78)
$I(DINS_NoEA_{i,t-30,t-3})$				-0.002 (-1.35)	-0.001 (-0.31)
$LaggedDependentVar_{i,t}$	0.158 (1.24)	0.158 (1.25)	0.157 (1.26)	0.156 (1.27)	0.380 (3.47)
$ForecastDispersion_{i,t}$	0.224 (1.14)	0.227 (1.15)	0.238 (1.20)	0.241 (1.22)	-0.374 (-11.94)
$\ln(MarketCapital)_{i,t}$	0.001 (3.45)	0.001 (3.77)	0.001 (3.65)	0.002 (3.87)	0.001 (2.03)
$\ln(MarketBook)_{i,t}$	0.001 (1.78)	0.001 (1.57)	0.002 (1.79)	0.001 (1.52)	0.000 (0.15)
$PastReturn_{i,t-30,t-3}$	0.005 (0.37)	0.004 (0.33)	0.004 (0.35)	0.004 (0.34)	0.09 (0.95)
# Obs.	3,621	3,621	3,621	3,621	1,077
Adj. R^2	6.57%	6.82%	6.64%	7.35%	25.41%

Table 7
The mechanisms: Author-track record and following

	Page View (1)	Read-to-End (2)
$Consistency_i$	49.151 (2.34)	27.754 (2.33)
$ArticleLength_i$	-27.663 (-1.94)	-20.523 (-2.31)
$NegSA_i$	-570.508 (-0.73)	-300.345 (-0.66)
$I(Blog_i)$	54.901 (3.80)	31.270 (3.28)
$I(Company_i)$	-24.978 (-1.33)	-12.598 (-1.19)
# Obs.	308	308
Adj. R^2	3.75%	4.18%

5.8 Table 7: Author track record and following

Read: Table 7 relates author attention to historical consistency. The coefficient on $Consistency_i$ is positive for both page views and read-to-end counts.

Economic message: Readers appear to reward authors with better track records. This gives authors an incentive to write useful research, because better historical performance is associated with more attention and potentially more compensation.

5.9 Table 8: Author track record and follower disagreement

Read: Table 8 shows that author consistency is negatively related to author-follower disagreement. The coefficient on $Consistency_i$ is -0.004 with a t -statistic of -2.22 .

Economic message: Commenters challenge weaker authors more than stronger authors. The comment section is not simply an echo chamber; it can act as a public evaluation mechanism.

Table 8
The mechanisms: Author-track record and follower disagreement

	Coefficient Estimate (<i>t</i> -statistic)
<i>Consistency_i</i>	-0.004 (-2.22)
<i>Article Length_i</i>	-0.002 (-2.47)
<i>NegSA_i</i>	0.240 (3.08)
<i>I(Blog_i)</i>	-0.000 (-0.12)
<i>I(Company_i)</i>	0.001 (0.94)
# Obs.	265
Adj. R^2	7.51%

Table 9
The mechanisms: Predictability when author-track record is poor and followers challenge the author

	(1)	(2)	(3)
<i>NegSA_{i,t}</i>	-0.013 (-0.03)	-0.687 (-1.21)	-0.664 (-1.16)
<i>NegSA-Comment_{i,t}</i>		-0.560 (-3.30)	-0.557 (-3.27)
<i>NegDJNS_{i,t}</i>			-0.611 (-1.15)
<i>I(DJNS_{i,t})</i>			0.019 (1.20)
<i>Upgrade_{i,t}</i>	0.017 (0.78)	0.015 (0.66)	0.014 (0.60)
<i>Downgrade_{i,t}</i>	0.013 (0.71)	0.012 (0.70)	0.012 (0.63)
<i>PosES_{i,t}</i>	-0.013 (-1.01)	-0.010 (-0.85)	-0.017 (-1.21)
<i>NegES_{i,t}</i>	-0.050 (-1.37)	-0.046 (-1.32)	-0.052 (-1.39)
<i>Volatility_{i,t}</i>	-0.026 (-0.86)	-0.026 (-0.88)	-0.026 (-0.84)
<i>ARet_{i,t}</i>	-0.099 (-0.76)	-0.099 (-0.76)	-0.109 (-0.83)
<i>ARet_{i,t-1}</i>	-0.123 (-0.64)	-0.124 (-0.65)	-0.123 (-0.65)
<i>ARet_{i,t-2}</i>	-0.128 (-0.90)	-0.136 (-0.97)	-0.132 (-0.94)
<i>ARet_{i,t-60,t-3}</i>	-0.017 (-0.55)	-0.016 (-0.54)	-0.017 (-0.56)
# Obs.	2,439	2,439	2,439
Adj. R^2	1.12%	1.48%	1.49%

5.10 Table 9: Poor author track records and crowd correction

Read: Table 9 restricts attention to cases where the author has a poor track record and comments disagree with the article. In this sample, comment negativity has a strong negative coefficient, while article negativity is not statistically reliable.

Economic message: When weak authors are challenged, the crowd signal is more reliable than the author signal. This is the paper's clearest evidence that comments can correct article-level mistakes.

6 Short summary

- The paper studies whether Seeking Alpha articles and comments predict stock returns and earnings surprises.
- Tone is measured as the fraction of negative words using a finance-oriented dictionary.
- The main return window is $t + 3$ to $t + 60$, which avoids overlap with the article and comment measurement window.
- More negative article tone predicts lower future abnormal returns.
- More negative comment tone also predicts lower future abnormal returns, even after controlling for article tone.
- The comment signal is stronger when it is based on more comments.
- Negative Seeking Alpha tone predicts negative earnings surprises, especially when articles are earnings-related.
- Authors with better past consistency receive more reader attention.
- Readers disagree more with historically weaker authors.
- When weak authors are challenged, the tone of comments predicts future returns more reliably than the tone of the article.

7 Conclusion

Chen, De, Hu, and Hwang show that social media can be a meaningful source of financial information. Seeking Alpha articles and comments predict future abnormal returns, and the same text signals also predict future earnings surprises. This second result is especially important because it suggests that social-media tone reflects fundamentals, not only investor attention or temporary price pressure.

The broader lesson is that peer-based investment advice can work when the platform creates enough structure: articles are stock-specific and substantive, comments create public feedback, readers can learn about author quality, and authors have reputational and monetary incentives to produce useful analysis. The paper therefore gives a finance-market version of the wisdom-of-crowds idea: dispersed investor opinions, when organized through a platform, can become informative.